

Dynamic Programming: Theory and Computation

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Topics

- DP problems
- DP theory
- DP algorithms

The focus of this lecture will be on

1. core theory
2. core algorithms and
3. numerical methods

For now we will stick to discrete problems

- finite action space A
- finite state space X

On the other hand, we will allow many interesting features

- Epstein-Zin preferences
- state-dependent discounting
- risk-sensitive preferences
- quantile preferences
- nonlinear discounting, etc.

Prelude: some linear algebra

In what follows we sometimes need to deal with equations of the form

$$v(x) = \sum_{x'} v(x') K(x, x') + b(x) \quad (x \in \mathbf{X})$$

Here K and b are given, while v is the unknown

We think of v as a function from finite set $\mathbf{X}$ to $\mathbb{R}$

Our objective is to solve the equation for v

To make this simple we can think of this in terms of matrices and vectors

Replacing $x = x_i$ with i , the equation

$$v(x) = \sum_{x'} v(x') K(x, x') + b(x) \quad \text{for } x \in \mathbf{X}$$

becomes

$$v_i = \sum_j v_j K_{ij} + b_i \quad \text{for } i \in \{1, \dots, n\}$$

or

$$v = Kv + b$$

The spectral radius of K is

$$r(K) := \max\{|\lambda| : \lambda \text{ is an eigenvalue of } K\}$$

```
import jax
import jax.numpy as jnp

@jax.jit
def spectral_radius(K):
    eigenvalues = jnp.linalg.eigvals(K)
    return jnp.max(jnp.abs(eigenvalues))
```

Theorem. If $r(K) < 1$, then

1. $I - K$ is invertible with

$$(I - K)^{-1} = \sum_{i \geq 0} K^i$$

2. and the equation $v = Kv + b$ has the unique solution

$$\begin{aligned} v^* &= (I - K)^{-1}b \\ &= \sum_{i \geq 0} K^i b \end{aligned}$$

Remark. This is the matrix analogue of

$$|a| < 1 \quad \implies \quad \sum_{i \geq 0} a^i = \frac{1}{1 - a}$$

Introduction to DPs: the Affine Case

Consider a dynamic program with

1. a finite set X called the **state space** and
2. a finite set A called the **action space**

Actions are restricted by a **feasible correspondence** Γ

- from X to A
- $\Gamma(x) =$ actions available in state x (nonempty)

The **feasible state-action pairs** is the set

$$G := \{(x, a) \in X \times A : a \in \Gamma(x)\}$$

The **Bellman equation** takes the form

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') K(x, a, x') \right\}$$

Components:

- the **reward function** r maps G to $\mathbb{R}$ and
- the **transition kernel** K maps $G \times X$ to $\mathbb{R}$

Example. Job search

Set up:

- wage offer process $(W_t)_{t \geq 0}$ is P -Markov on finite set W :

$$\mathbb{P}\{W_{t+1} = w' \mid W_t = w\} = P(w, w')$$

- discount factor $\beta \in (0, 1)$

The Bellman equation is

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{w'} v(w') P(w, w') \right\}$$

We can rewrite the Bellman equation as

$$v(w) = \max_{a \in \{0,1\}} \left\{ a \frac{w}{1-\beta} + (1-a) \left[c + \beta \sum_{w'} v(w') P(w, w') \right] \right\}$$

This is an affine DP with

- state $w \in W$
- $\Gamma(w) = \{0, 1\}$
- $r(w, a) = aw/(1-\beta) + (1-a)c$
- $K(w, a, w') = (1-a)\beta P(w, w')$

Example. Firm exit

Firm observes current state of the world z and decides whether to exit

- If exit then receive scrap value $s(z)$
- If continue then receive profit $\pi(z)$

Set up:

- state of the world $(Z_t)_{t \geq 0}$ is P -Markov on finite set Z
- discount factor $\beta = 1/(1 + r)$ with $r > 0$

The Bellman equation is

$$v(z) = \max \left\{ s(z), \pi(z) + \beta \sum_{z'} v(z') P(z, z') \right\}$$

We can rewrite the Bellman equation as

$$v(z) = \max_{a \in \{0,1\}} \left\{ as(z) + (1-a) \left[\pi(z) + \beta \sum_{z'} v(z') P(z, z') \right] \right\}$$

This is also an affine DP

- $r(z, a) = as(z) + (1-a)\pi(z)$
- $K(z, a, z') = (1-a)\beta P(z, z')$

One limitation: constant discounting

For this model we would expect

$$\beta = \frac{1}{1 + \text{interest rate}}$$

And interest rates vary over time

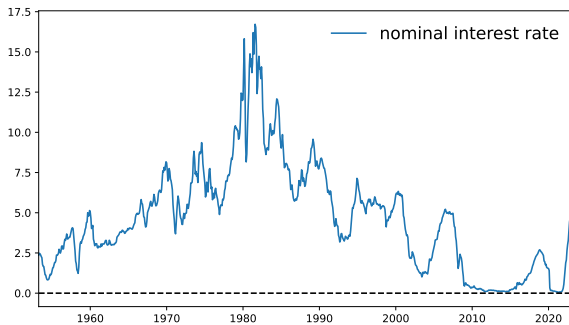


Figure: Nominal US interest rates

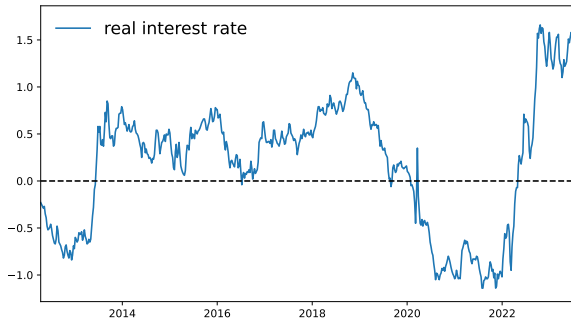


Figure: Real US interest rates

Example. Firm exit with state-dependent discounting

The Bellman equation is

$$v(z) = \max_{a \in \{0,1\}} \left\{ as(z) + (1-a) \left[\pi(z) + \beta(z) \sum_{z'} v(z') P(z, z') \right] \right\}$$

This is also an affine DP

- $r(z, a) = as(z) + (1-a)\pi(z)$
- $K(z, a, z') = (1-a)\beta(z)P(z, z')$

Example. A **shortest path problem** on graph $\mathcal{G} = (\mathbf{X}, E)$

- $c(x, x') = \text{cost of traversing edge } (x, x') \in E$
- the direct successors of x denoted by

$$\mathcal{O}(x) := \{x' \in \mathbf{X} : (x, x') \in E\}$$

Aim: find the minimum cost path from x to a specified vertex d

The Bellman equation is

$$v(x) = \min_{x' \in \mathcal{O}(x)} \{c(x, x') + v(x')\}$$

This is an affine DP with

- state $x \in \mathbf{X}$ and action $a \in \mathbf{X}$
- $\Gamma(x) = \mathcal{O}(x)$
- $r(x, a) = -c(x, a)$
- $K(x, a, x') = \mathbb{1}\{x' = a\}$

Example. Inventory management

Bellman equation is

$$v(y) = \max_{a \in \Gamma(y)} \left\{ r(y, a) + \beta \sum_{y'} v(y') R(y, a, y') \right\}$$

- $y \in \{0, \dots, \bar{y}\}$ is current inventory
- a is current order (units of stock)
- $r(y, a)$ is current profits
- $R(y, a, y') = \text{prob. next state is } y' \text{ given } y, a$
- An affine DP with $K(y, a, y') = \beta R(y, a, y')$

Let's now return to the general case with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') K(x, a, x') \right\}$$

- state space is X (finite)
- action space is A (finite)
- reward function r maps G to $\mathbb{R}$
- transition kernel K maps $G \times X$ to $\mathbb{R}$

Definitions

A **feasible policy** is a map $\sigma: X \rightarrow A$ with

$$\sigma(x) \in \Gamma(x) \quad \text{for all } x \in X$$

- $\Sigma :=$ all feasible policies

Feasible policy σ is called ***v-greedy*** if

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') K(x, a, x') \right\} \quad \forall x \in X$$

The **Bellman operator** is

$$(Tv)(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') K(x, a, x') \right\}$$

For each $\sigma \in \Sigma$, we introduce the **policy operator**

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \sum_{x'} v(x') K(x, \sigma(x), x')$$

Note:

$$\sigma \text{ is } v\text{-greedy} \iff Tv = T_\sigma v$$

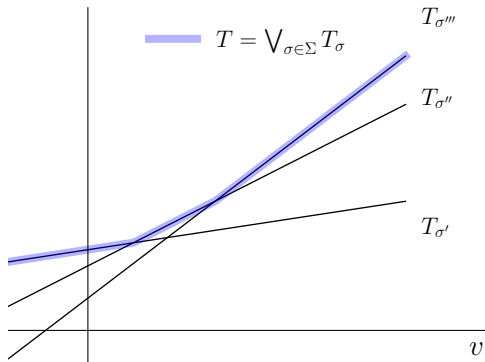


Figure: T is the pointwise max. of $\{T_{\sigma}\}_{\sigma \in \Sigma}$

Add an intuitive discussion for the recursion that defines v_σ in the MDP case

Let

- $r_\sigma(x) := r(x, \sigma(x)) = \text{rewards under } \sigma$
- $K_\sigma(x, x') := K(x, \sigma(x), x') = \text{transitions under } \sigma$

The **lifetime value function** associated with policy σ is

$$v_\sigma := (I - K_\sigma)^{-1} r_\sigma$$

Note that

$$v \in \text{fix}(T_\sigma) \iff v = r_\sigma + K_\sigma v$$

$$\iff (I - K_\sigma)v = r_\sigma$$

$$\iff v = (I - K_\sigma)^{-1} r_\sigma =: v_\sigma$$

Defining optimality

We define the **value function** via

$$v^*(x) := \max_{\sigma \in \Sigma} v_{\sigma}(x) \quad (x \in \mathbf{X})$$

Equivalently,

$$v^* := \bigvee_{\sigma} v_{\sigma}$$

A policy σ is called **optimal** if $v_{\sigma} = v^*$

MDP Optimality

Theorem. For an MDP with Bellman operator T and value function v^* ,

1. v^* is the unique fixed point of T in $\mathbb{R}^X$
2. T is a contraction mapping on $\mathbb{R}^X$
3. A feasible policy is optimal if and only if it is v^* -greedy
4. At least one optimal policy exists

Standard algorithms

Algorithm 1: VFI

input $v_0 \in \mathbb{R}^X$

input τ , a tolerance level for error

$\varepsilon \leftarrow +\infty$

$k \leftarrow 0$

while $\varepsilon > \tau$ **do**

$v_{k+1} \leftarrow Tv_k$

$\varepsilon \leftarrow \|v_k - v_{k+1}\|_\infty$

$k \leftarrow k + 1$

end

Compute a v_k -greedy policy σ

return σ

Algorithm 2: HPI

input $\sigma_0 \in \Sigma$, set $k \leftarrow 0$ and $\varepsilon \leftarrow 1$

while $\varepsilon > 0$ **do**

$v_k \leftarrow$ the lifetime value of σ_k

$\sigma_{k+1} \leftarrow$ a v_k -greedy policy

$\varepsilon \leftarrow \mathbb{1}\{\sigma_k \neq \sigma_{k+1}\}$

$k \leftarrow k + 1$

end

return σ_k

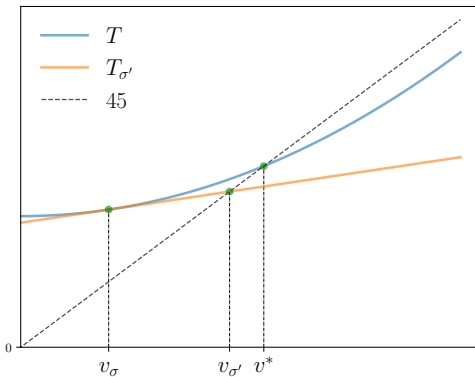


Figure: HPI as a version of Newton's method

Algorithm 3: OPI

input v_0 , an initial guess of v^*

input τ , a tolerance level for error

input $m \in \mathbb{N}$, a step size

$k \leftarrow 0$

$\varepsilon \leftarrow +\infty$

while $\varepsilon > \tau$ **do**

$\sigma_k \leftarrow$ a v_k -greedy policy

$v_{k+1} \leftarrow T_{\sigma_k}^m v_k$

$\varepsilon \leftarrow \|v_k - v_{k+1}\|_\infty$

$k \leftarrow k + 1$

end

return σ_k

Proposition. Under the stated condition, VFI, HPI and OPI all converge

Moreover, HPI converges to an exact optimal policy in finitely many steps

For details and proofs see Ch. 5 of <https://dp.quantecon.org/>